

Dataset: retail_dataset

Python:

- Data cleaning:
 1. Missing values are checked and those in categorical columns are replaced with 'Not Provided' to avoid any misconceptions:

```
missing_data = df.isnull().sum()
print("Missing Data:")
missing_data
```

Customer_ID	308	Time	350
Name	382	Total_Purchases	361
Email	347	Amount	357
Phone	362	Total_Amount	350
Address	315	Product_Category	283
City	248	Product_Brand	281
State	281	Product_Type	0
Zipcode	340	Feedback	184
Country	271	Shipping_Method	337
Age	173	Payment_Method	297
Gender	317	Order_Status	235
Income	290	Ratings	184
Customer_Segment	215	products	0
Date	359		
Year	350		
Month	273		

2. Duplicated values are checked and dropped:

```
duplicates = df.duplicated()
df[duplicates]
```

```
df.drop_duplicates(inplace=True)
```

3. Missing values in numerical values are replaced with Median:

```
df['Total_Purchases'].fillna(df['Total_Purchases'].median(), inplace=True)
df['Amount'].fillna(df['Amount'].median(), inplace=True)
df['Total_Amount'].fillna(df['Total_Amount'].median(), inplace=True)
```

4. Data types of some categories are changed:

```
df["Time"] = pd.to_datetime(df["Time"], format="%H:%M:%S", errors="coerce")

df["Hour"] = df["Time"].dt.hour

df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
```

```
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
```

```
df["Ratings"] = pd.to_numeric(df["Ratings"], errors="coerce")
```

5. New columns are created to extract useful information:

```
#Extract data separately
df["Year"] = df["Date"].dt.year
df["Month"] = df["Date"].dt.month
df["Quarter"] = df["Date"].dt.quarter
```

MySQL results:

```
1 • SELECT
2     COUNT(*) AS total_customers,
3     AVG(Age) AS avg_age,
4     AVG(Total_Purchases) AS avg_purchases,
5     AVG(Total_Amount) AS avg_spending
6 FROM final_retail_data;
```

result Grid  Filter Rows: Export:  Wrap Cell Conte			
total_customers	avg_age	avg_purchases	avg_spending
86	44.24418604651163	5.2558	1377.3335727267438

What is the overall profile and purchasing behavior of the customer base?

The dataset consists of 86 customer observations. The average customer age is approximately 44 years, indicating a predominantly middle-aged customer base. Customers make an average of just over five purchases, suggesting moderate repeat purchasing behavior. Mean total spending per customer is approximately 1,377 units, reflecting a relatively high customer value.

```
15 • SELECT Year, Month, SUM(Amount) AS monthly_sales
16 FROM final_retail_data
17 GROUP BY Year, Month
18 ORDER BY Year, Month;
```

Result Grid Filter Rows: Export: Wrap Cells			
Year	Month	monthly_sales	
2023	3	2832.2052936	
2023	4	2277.15934568	
2023	5	2148.2594562	
2023	6	2267.29123674	
2023	7	2151.60097711	
2023	8	1071.5194934	
2023	9	1623.13826973	
2023	10	1812.8416676200002	
2023	11	1088.55071246	
2023	12	1971.27207641	
2024	1	3045.3905757999996	
2024	2	677.78574809	

How does total revenue evolve over time on a monthly basis?

Monthly sales analysis reveals relatively stable performance throughout mid-2023, followed by a noticeable decline in late summer. Sales recovered toward the end of the year, with a pronounced peak observed in January 2024. The sharp decline in February 2024 is likely attributable to partial data coverage and should be dealt with cautiously.

```

21 • SELECT Product_Category, COUNT(*) AS total_orders, SUM(Amount) AS total_sales
22 FROM final_retail_data
23 GROUP BY Product_Category
24 ORDER BY total_sales DESC;
25

```

Product_Category	total_orders	total_sales
Books	18	5717.470761129999
Grocery	21	5192.06216885
Electronics	16	4868.07783305
Clothing	15	4062.1039872599995
Home Decor	16	3127.3001025500007

Which product categories generate the most orders and revenue?

Product category analysis indicates that Books generate the highest total sales despite a lower order volume than Grocery, suggesting higher average order values. Grocery exhibits the highest transaction frequency but lower revenue per order, while Electronics also demonstrates high-value purchasing behavior. Home Decor contributes the least to total sales, reflecting lower average transaction values.

```

27 • SELECT Customer_Segment, COUNT(*) AS customers, AVG(Total_Amount) AS avg_spent
28 FROM final_retail_data
29 GROUP BY Customer_Segment;
30

```

Customer_Segment	customers	avg_spent
Regular	85	1386.1243759929407
New	1	630.1152951

How does spending behavior differ across customer segments?

Customer segmentation analysis reveals that regular customers account for the vast majority of observations and demonstrate significantly higher average spending compared to new customers.

```
12 • SELECT Payment_Method, COUNT(*) AS usage_count FROM final_retail_data GROUP BY Payment_Method;
```

Payment_Method	usage_count
Debit Card	23
Cash	19
PayPal	20
Credit Card	24

What payment methods do customers prefer?

Analysis of payment methods indicates a relatively balanced distribution of usage across credit cards, debit cards, PayPal, and cash. Electronic payment methods slightly dominate overall transactions, while cash payments remain a relevant option.

```
35 • SELECT Shipping_Method, COUNT(*) FROM final_retail_data GROUP BY Shipping_Method;
36
```

Shipping_Method	COUNT(*)
Same-Day	28
Standard	30
Express	28

What shipping options do customers most frequently choose?

Analysis of shipping method selection reveals a balanced distribution across standard, same-day, and express delivery options. The absence of a dominant shipping method suggests that customers prioritize flexibility and context-specific delivery choices.

```

38 • SELECT
39     AVG(Ratings) AS avg_rating,
40     COUNT(CASE WHEN Feedback IS NOT NULL THEN 1 END) AS feedback_count
41     FROM final_retail_data;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
avg_rating	feedback_count			
2.802325581395349	86			

What is the overall level of customer satisfaction and feedback availability?

The overall average customer rating is approximately 2.8, which is below the neutral midpoint of the rating scale, suggesting moderate dissatisfaction among customers. Feedback was available for all records, ensuring comprehensive coverage of customer experience data.

```

72 • SELECT
73     AVG(
74         CASE
75             WHEN Income = 'low' THEN 1
76             WHEN Income = 'medium' THEN 2
77             WHEN Income = 'high' THEN 3
78         END
79     ) AS avg_income_level
80     FROM final_retail_data;

```

Result Grid		Filter Rows:	Export:
avg_income_level			
2.0814			

What is the overall income profile of the customer base?

Income was treated as an ordinal categorical variable and encoded numerically to compute an average income level index. The resulting average value of approximately 2.08 indicates that the customer base is centered around a medium income level, with a slight skew toward higher-income customers.

```
101 • SELECT
102     Products,
103     COUNT(*) AS total_orders,
104     SUM(Amount) AS total_sales
105 FROM final_retail_data
106 GROUP BY Products
107 ORDER BY total_sales DESC
108 LIMIT 10;
```

Products	total_orders	total_sales
Nokia	2	890.2667007
Loafers	2	669.4753979
Travel	2	658.6864112000001
Chocolate bars	2	636.2976021
LCD TV	2	540.87496126
Art supplies	1	485.0773617
TV stand	1	477.7545633
History	1	474.8269416
Mineral water	1	472.6264585
Literary fiction	1	466.22066

Which individual products generate the highest total revenue?

Product-level analysis reveals a high degree of revenue concentration, with the top-selling products generating substantial revenue from only one or two orders. This indicates that overall sales performance is strongly influenced by high-value transactions rather than purchase frequency. The top-performing products span multiple categories, particularly electronics and books, highlighting the importance of premium and high-ticket items.

```

122 • SELECT
123     City,
124     State,
125     SUM(Amount) AS total_sales
126 FROM final_retail_data
127 GROUP BY City, State
128 ORDER BY total_sales DESC
129 LIMIT 10;

```

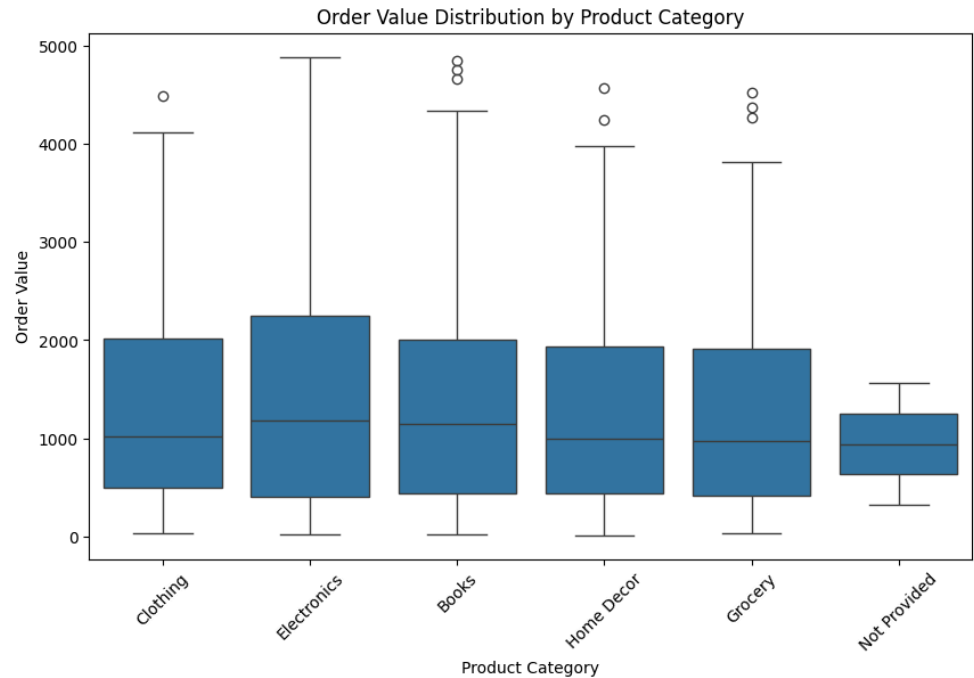
Result Grid			
Filter Rows:			
	City	State	total_sales
	Portsmouth	England	22543.928448540002
	Kitchener	Ontario	315.0576476
	Dortmund	Berlin	108.0287567

Where does revenue originate geographically?

City sales analysis reveals a high degree of geographic concentration, with Portsmouth accounting for the largest share of total sales. Other cities such as Kitchener and Dortmund contribute comparatively smaller proportions, indicating uneven regional market penetration.

The analyses show that retail performance is shaped by **seasonal demand**, **diversified product categories**, **repeat customer dominance**, **high-value individual products**, and **geographically concentrated markets**. While revenue performance is stable and diversified, **moderate customer satisfaction scores** suggest that improving customer experience represents a key opportunity for growth.

Charts with Python:

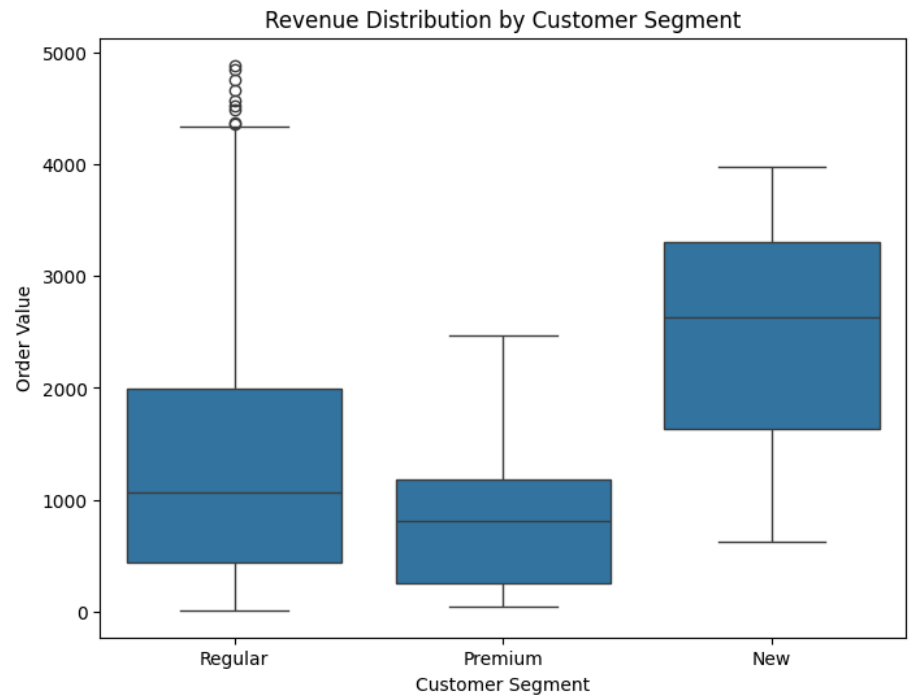


How does order value vary across product categories?

Which categories have higher typical order values?

Which categories show greater variability and outliers?

The box plot reveals substantial variation in order values across product categories. Electronics and Books exhibit the highest variability and the largest number of extreme outliers, indicating occasional high-value purchases. Grocery and Clothing show more stable and predictable order values, reflecting volume-driven purchasing behavior. The presence of extreme outliers in several categories suggests that a small number of transactions contribute disproportionately to total revenue.



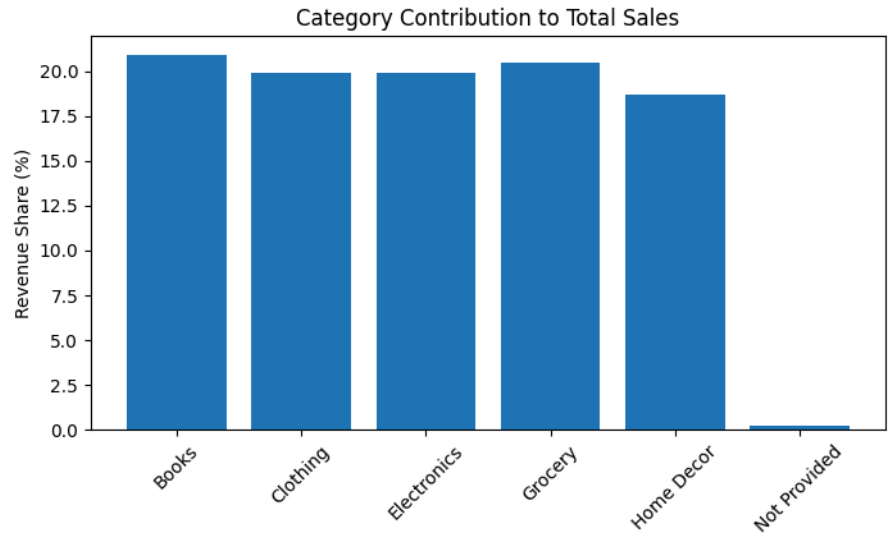
How does spending behavior differ between customer segments?

Which segment places higher-value orders?

Which segment shows more volatile spending?

Do new customers behave differently from regular or premium ones?

Order value distributions vary considerably across customer segments. New customers exhibit the highest median order value, suggesting strong initial purchases, while regular customers display the greatest variability and the highest extreme values. Premium customers show more stable but comparatively lower per-order spending. These results highlight the importance of distinguishing between transaction value and long-term customer value.



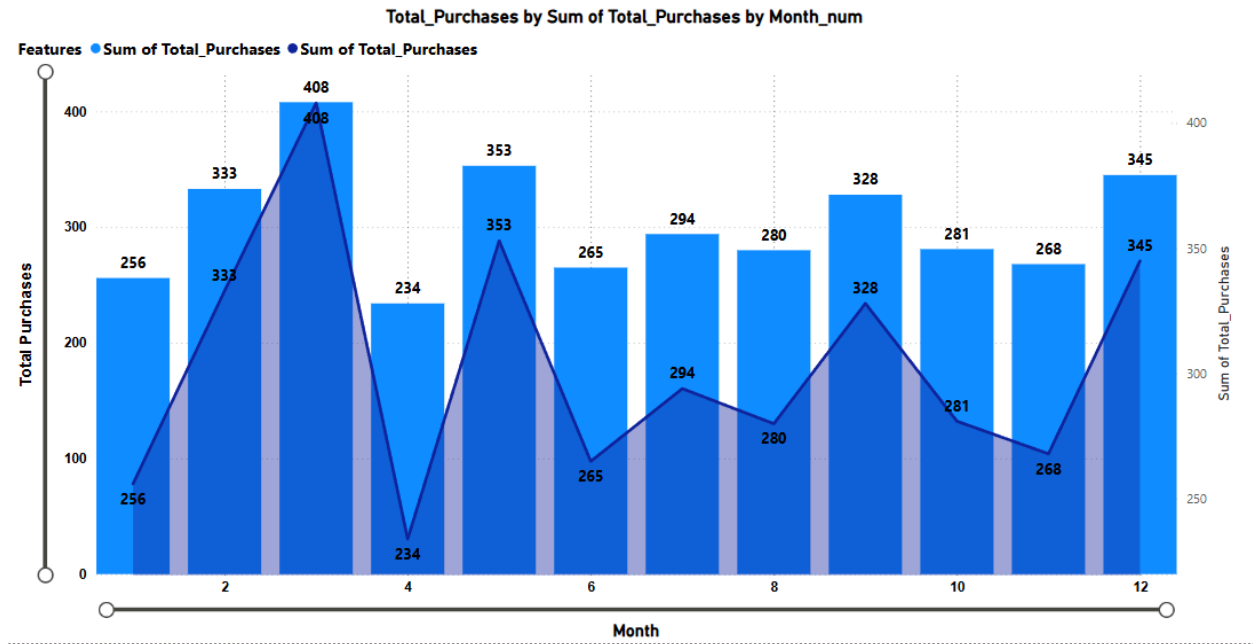
Which product categories contribute most to total revenue?

Is revenue concentrated or diversified across categories?

Are there categories with negligible impact?

Revenue contribution analysis shows a balanced distribution across major product categories, with Books, Grocery, Electronics, and Clothing each contributing approximately one-fifth of total sales. This indicates a diversified revenue structure, reducing dependency on any single category. The negligible contribution of uncategorized transactions suggests strong data completeness.

Charts with Power BI:

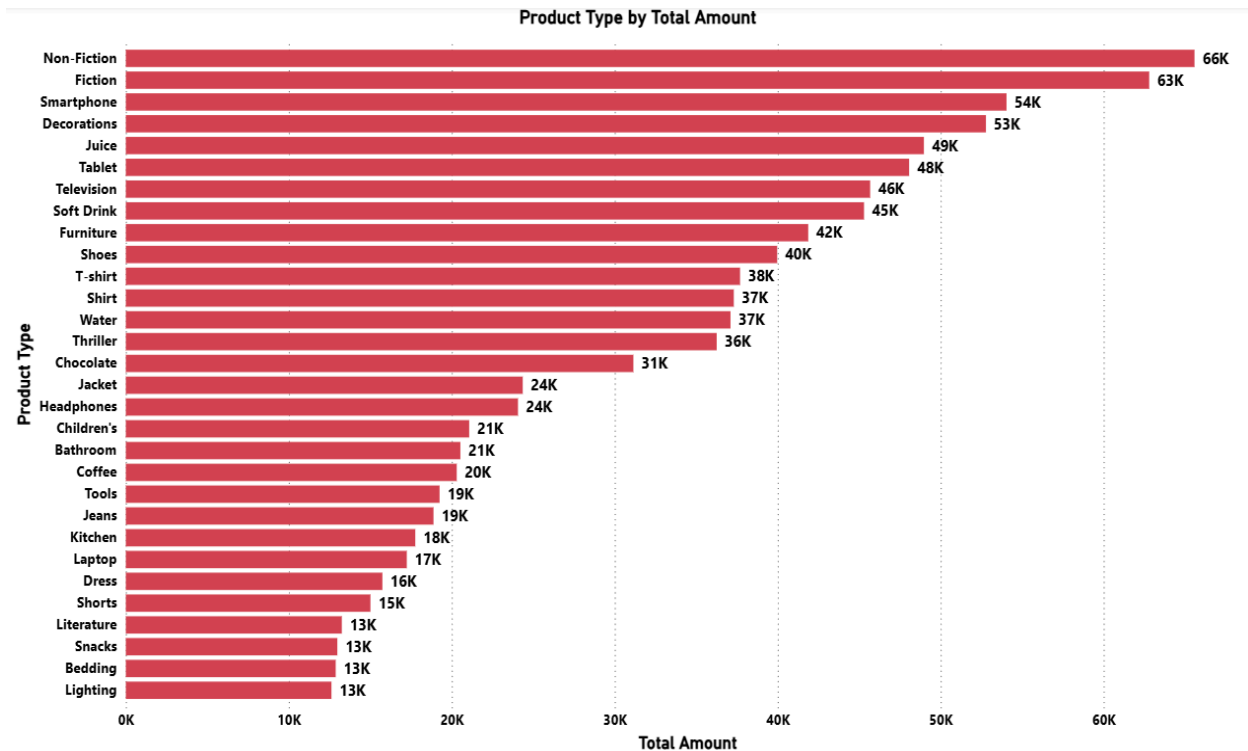


How does customer purchasing activity change over time?

Are there seasonal peaks or drops in purchases?

Which months have the highest and lowest transaction volume?

Monthly purchase volume shows pronounced seasonal variation, with a major peak in March followed by a temporary decline in April and a rapid recovery in May. Purchasing activity remains relatively stable throughout the summer months and increases again toward the end of the year, culminating in a strong December peak. This pattern suggests seasonality rather than structural volatility in customer purchasing behavior.

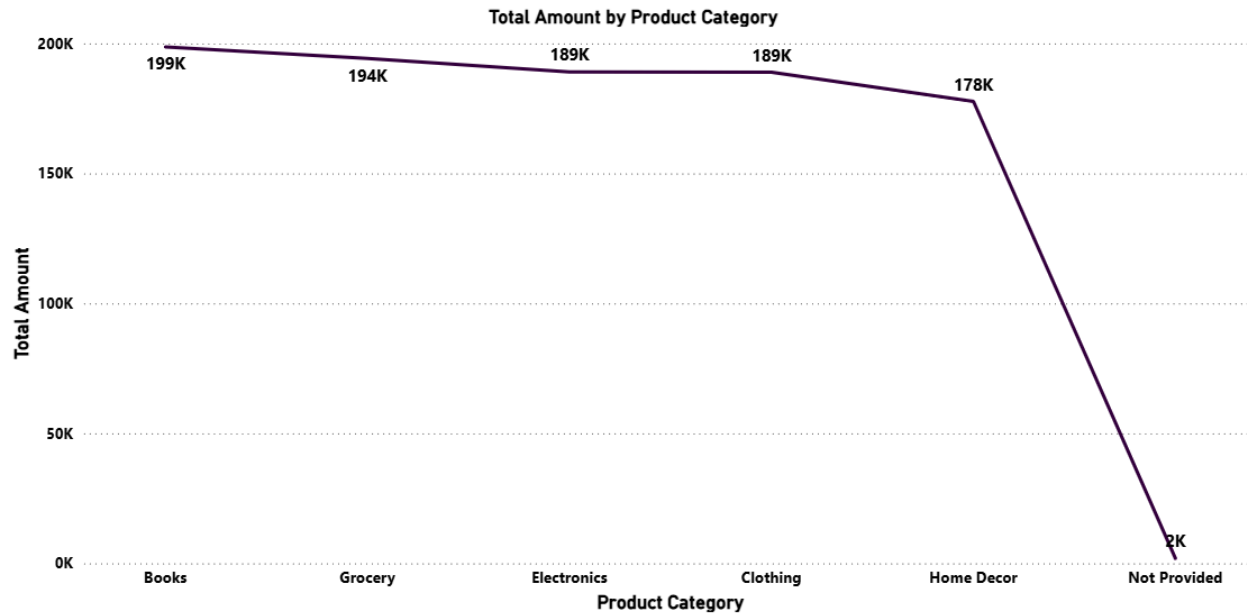


Which individual product types generate the most revenue?

Is revenue driven by many products or a few top performers?

Does a Pareto / long-tail pattern exist?

Product-level revenue analysis reveals a highly skewed distribution, with Non-Fiction and Fiction books generating the highest total revenue, followed by high-value electronics such as smartphones and tablets. Consumable goods contribute through frequent purchases, while apparel and home products exhibit more evenly distributed revenue across multiple items. These findings indicate a diversified revenue structure driven by distinct purchasing mechanisms across product types.

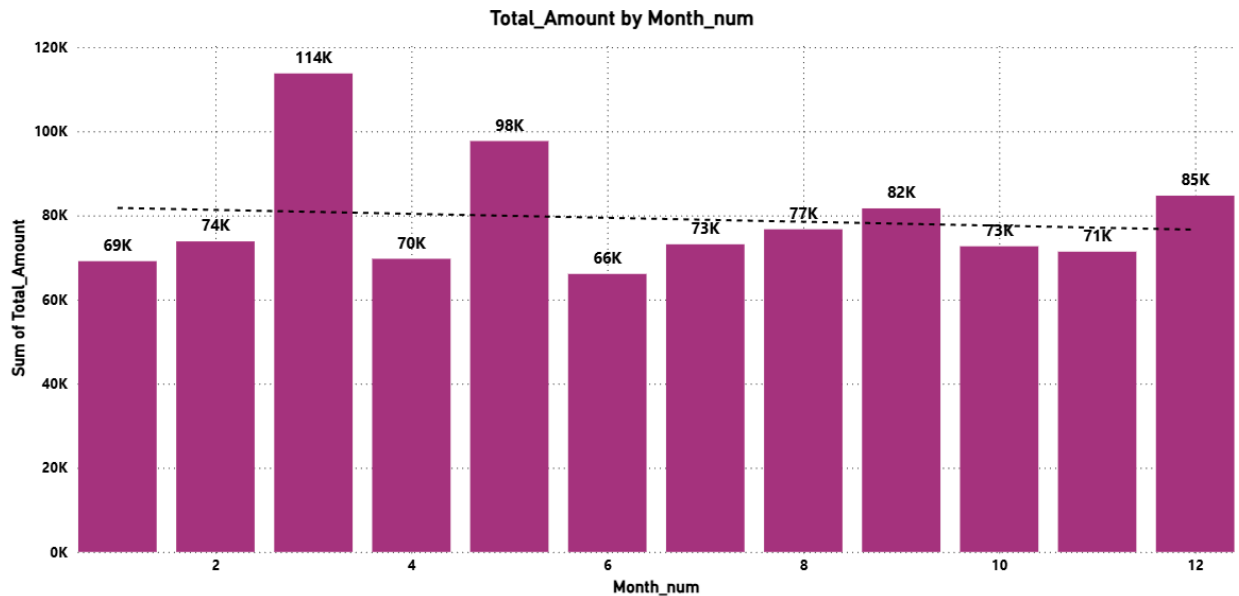


How do total revenues compare across categories?

Are category revenues similar or sharply different?

Is any category dominant or underperforming?

Total revenue is distributed relatively evenly across the main product categories, with Books and Grocery contributing the largest shares, followed closely by Electronics and Clothing. Home Decor contributes slightly less, while uncategorized transactions account for a negligible proportion of total revenue. The data represent categorical comparisons rather than a temporal trend.



How does revenue change month by month?

Are there revenue peaks and slow periods?

Is overall revenue stable or volatile?

How does monthly revenue compare to the average benchmark?

Monthly revenue analysis reveals pronounced seasonality, with a major peak observed in March and secondary peaks in May and December. Revenue reaches its lowest point in June, followed by a steady recovery toward the end of the year. Overall, revenue fluctuates around a stable average, indicating sustained performance rather than structural decline.

All in all, the analyses show that the business benefits from **diversified revenue streams, clear seasonality, strong product-level performers, and distinct customer behaviors**, but also faces challenges related to **customer satisfaction and uneven geographic concentration**. Revenue is driven by different mechanisms—high-value items, frequent consumables, and loyal repeat customers—indicating that **segmented and category-specific strategies are essential**.